

EAC GALLERY



MUT RESEARCH AWARDS

RESEARCH AWARDS SHOW GROWTH IN THE RESEARCH TRAJECTORY

As MUT held its 20th annual Research Awards, Acting Vice-Chancellor and Principal of MUT, Professor Marcus Ramogale noted how this event marks MUT's coming of age in research. Moving from an institution with less than 10 PhDs in the year 2000, the University has grown and is active in "I would like to remind you that last year, MUT was recognised by the National Research Foundation (NRF) for its upward trajectory and quality of research output. As a result, MUT received an Exceleration award, a portmanteau word created by the NRF from 'excellence' and 'acceleration'. This has marked the cornerstone of our research agenda and there is no more turning back", Professor Ramogale said. He acknowledged the work that Dr Anette Minnie, MUT's research director had accomplished to get MUT's research recognised at this level.

"I am therefore heartened that the University has now appointed Professor Nokuthula Sibiyi as a Deputy Vice-Chancellor for Research, Innovation and



Professor Marcus Ramogale

Engagements to move MUT research to another level", Professor Ramogale said, and acknowledged his faith that Professor Sibiyi will deliver on improving the University's research. Professor Ramogale described the situation as "chalk and cheese", and that there was "more coming our way. We are here because of our researchers. We urge you to do more," he said. Professor Ramogale also said that the University's research output was commended by the Department of Higher Education and Training (DHET).

WATCH THIS SPACE, WE ARE HERE TO SHAKE THE MUT RESEARCH AGENDA

At the MUT glitz and glamour research awards evening, the newly appointed Research, Innovation and Engagement (RIE) Deputy Vice-Chancellor, Professor Nokuthula Sibiyi, spoke positively about headway that MUT had done with research. "I was very impressed about the milestones accomplished at this time of the year already, and the strategic direction being taken. The University's research output is on the up. I am humbled to be joining this MUT family and I will walk with all of you to take MUT to the next level", she said.

Professor Sibiyi emphasised the fact that MUT had to use research to contribute to its third-stream income. She advised the University staff to join hands, and work together so they can meet their target. "One of my first 'to do' programmes would be to convene a Research Indaba which will carve a strategic direction for the research agenda", she said.

In less than two weeks of her coming into office, Professor Sibiyi had already undertaken an audit of MUT's research. "I am pleased, and I have an idea of how I will assimilate myself into the groundwork that has been paved for me. I wish to assure you that I have



Professor Nokuthula Sibiyi

also contributed to the University's research output by publishing a paper under MUT. More is still to come. I come here as a partner who will also make a meaningful contribution to growing MUT's research", she said.

Speaking on behalf of the researchers, Dr Taurai Mutanda assured Professor Sibiyi that they would be working with her to change MUT's research output.

ROCKSTAR PROFESSOR WITH A SALT AND PEPPER OUTLOOK SCOOPS THE TOP AWARD

Professor Babatunde Bakare, the Acting Dean of the Faculty of Engineering, was recognised by MUT's Senate as the most suitable researcher qualifying for a Senate Prize at MUT's Research Awards. As the highest decision-making body, Senate agreed that the award this year should go to no other than Professor Bakare, known affectionately as Tunde to his friends.

At only 43 years old, Professor Bakare has gained stature as a prolific researcher not just at MUT but globally. Last year, Professor Bakare published 10 journal articles, and contributed two book chapters qualifying him to win the coveted Senate Award. This is the second time in a row that Professor Bakare has won the Senate



Professor Babatunde Bakare scoops a prize. Here he is with Professor Noks Sibiyi

Prize. He is a well-respected researcher both at home and internationally.

In total he has published 45 research papers and mentored and supervised four PhDs. Professor Bakare collaborates with universities in Africa, USA, and the UK.

He has published in high impact factor journals.

He is married with two children, yet he still finds time to not only dedicate on research but also on empowering others. Well done to Professor Bakare.

THE ENGINEERING FACULTY SHINES AT THE AWARDS

Professor Bakare's leadership and his empowerment agenda has been cascaded down to the Engineering faculty which he leads. At the research awards ceremony, three staff members from his faculty also received awards. They are Dr Kaniki Tumba, lecturer in the department of chemical engineering who won the Most Productive Researcher in the faculty and Dr Papy Numbi, a lecturer in the department of electrical engineering who was the runner-up in the same category. The Most Productive Novice Researcher Award went to Dr Joseph Bwapwa, a lecturer in the department of civil engineering.

Dr Tumba's research focus spans from chemical engineering thermodynamics to separation processes, bioprocessing and renewable energy. "My other achievements include the membership to the editorial board of two reputable journals, which are the South African Journal of Chemical Engineering and The Open Chemistry Journal," he said. Over the last seven years, Dr Tumba has co-authored publications with researchers from various countries, including Algeria, the Democratic Republic of Congo, India, Iran, Germany, France, China, and Finland. "This represents a good foundation for fruitful and well-structured collaborations that will benefit MUT research and the internationalisation agenda," he said.

Dr Numbi's research focuses on renewable energy, energy efficiency and management. He is particularly interested in utilising optimisation techniques to address energy problems in industrial sectors with heavy energy demands, such as mining. He holds a Y rating from the NRF and is registered with several professional engineering bodies.

Dr Bwapwa, a senior lecturer in the Department of Civil Engineering focuses his research on water and wastewater treatment remediation. This is also coupled with expertise in green fuels with low carbon emission



Dr Papy Numbi



Dr Joseph Bwapwa, with Professor Nok Sibiya

from biomass and wastes. He has a good track record of publications in high impact factor journals and has published in peer reviewed journals and presents at international conferences. He is well known for chairing a few sessions in international conferences. He is also a mentor to junior academic staff doing research.

MUT STUDY IDENTIFIES GEOENVIRONMENTAL FACTORS AS POSSIBLE CAUSES OF DISEASES OF UNKNOWN AETIOLOGY



Professor Theophilus Davies

Despite the enormous advances made by modern medicine, there are several diseases today whose causes are still unknown. These diseases are generally referred to as diseases of unknown aetiology (DUA), and as recent as 2020, an estimated 76% of unknown disease outbreaks remained undiagnosed." This is according to a recent study by renowned Medical Geologist and Research Professor in the Faculty of Natural Sciences at Mangosuthu University of Technology (MUT), Professor Theophilus Davies. The study, titled, "*The position of geochemical variables as causal co-factors of diseases of unknown aetiology*", has called for greater consideration to be given to the contribution of geoenvironmental factors in a multi-factor explanation and diagnosis of DUA.

Published in the *SN Applied Sciences – A Springer Nature Journal (4:236)*, Professor Davies' study comes at the back of a 2018 World Health Organization study on worldwide cancer mortality which identified 10 diverse environmental risk factors, including "some with links to the geological environment, such as air pollution and ionizing radiation exposure".

What makes Professor Davies' study novel, and groundbreaking is that it expands the scope of explanation of what causes diseases of unknown origins and their diagnoses to include geoenvironmental factors by drawing on his research expertise in Medical Geology, which Professor Davies co-founded.

In the study, Professor Davies explained that it was important to bring Medical Geology expertise because this field of study is instrumental in providing credible explanations on the "exposure and transfer mechanisms

of geochemicals to the food-chain and humans", along with "the nature of related ecotoxicological and health effects that are produced".

Professor Davies' study identified eight geoenvironmental and related co-factors that can significantly contribute to the resolution of causation of DUA if they are properly considered.

These factors include: "the immune-modulatory effect of geochemical variables that underline nutritional and potentially toxic element perturbations in metabolic processes"; damage to DNA along with the synthesis of reactive oxygen species caused by the imbalance of trace elements/metals and/or metalloids in the metabolism; air, soil and water pollution from a diversity of sources; "geogenic dust particles from mining, ore processing and vehicular transportation on untarred roads"; over-exposure to radiation and radionuclides in soil, water and air during mining and ore processing; seasonal variations and geographical patterns; "climate change and geoclimatic effects"; and "factors of geopathic stress and heat stress."

The study also laid out details of various factors that impact the link between geoenvironmental factors and diseases. These include the geochemical variables and diseases; geochemical variables and the immune system; and disease risk mapping.

"In order to promote immune-immediate health for life, we must consider the importance of our exposure to geoenvironmental variables and the dynamics of pathogen invasion in immune programming," concluded Professor Davies.



A JOURNEY TOWARDS INCREASING MUT'S RESEARCH OUTPUT

A quick assessment by the recently appointed Deputy Vice-Chancellor, Research, Innovation and Engagement (DVC: RIE), Professor Nokuthula Sibiya led her to convening a Research Indaba to find how synergies among departments could increase MUT's research output. With three portfolios: research, innovation and engagements linked within her portfolio; Professor Sibiya believes that this provides fertile ground for MUT's research to thrive.

Professor Sibiya urged staff to take advantage of the resources that are at their disposal within MUT. "The most important of these resources are the Technology Station in Chemicals (TSC), and the Community Engagement and Development (CEAD) directorates", she said. In addition, Professor Sibiya has roped in the Library Services as a strategic stakeholder towards strengthening the new research trajectory.

Her approach as outlined during the Indaba shows a helicopter view or clarity of purpose approach; where all enabling departments reinvigorate their commitment to provide an enabling environment for MUT's research to accelerate. For instance, the staff that research water and chemicals, and staff that are involved in student



Professor Nokuthula Sibiya, right, with Dr Anette Mienie, Director: Research

entrepreneurship can benefit from working with the TSC. While presenting the TSC's work, Xolile Ngubane, Acting Director of TSC informed staff that the TSC trained Small, Medium and Micro Enterprises (SMMEs) on how to manufacture their products and make them available to the market. The marketing part is what



The MUT staff that attended the event



Professor Busisiwe Nkonki,
Director of CEAD Mandleni

staff and students involved in entrepreneurship should know. Staff researching renewable energy, and water studies can also benefit from working with the TSC.

The Director of CEAD, Professor Busisiwe Nkonki-Mandleni, highlighted the benefits of having community engagement projects registered with her division.

She said that once community engagement projects are registered with her directorate, project owners could rely upon her division to support them when they apply for a promotion. Professor Nkonki-Mandleni emphasised that community engagement was as important as research and teaching. It is one of the three pillars of the university, she said. Professor Nkonki-Mandleni also stressed the need for staff to understand what 'engage' meant by saying "We need to be an engaged university, and know what this means".

During her presentation, Dr Nthabiseng Mosala-Bryant, Senior Director of Library Services, informed staff about the importance of utilising the library to its fullest, along with publishing their research papers in reputable journals.

Dr Anette Mienie, the Director of Research directorate, said Professor Sibiya has proposed the appointment of a committee, which will look at all the areas of concern, and deal with tying up all the loose ends.



Dr Nthabiseng Mosala-Bryant,
Senior Director of Library Services



MUT HOSTS WORLD- RENOWNED ENVIRONMENTALIST

T rue to the accolades by his peers across the globe, Dr Hilary Inyang did not disappoint as a guest speaker during MUT's public lecture on "Greening of the Economy for Livelihood Improvement in Africa as a solution in addressing Energy, Water and Food Sustainability". He brought a research rigor not seen in the country for a while. Described in various media as an expeditionist and educator in the areas of geoenvironmental science and engineering; the word, 'world renowned' is no misnomer when it comes to describing what he knows, stands for, and has practiced.

Dr Inyang's visit to MUT was coordinated by MUT's research directorate through engagements with the Faculty of Natural Sciences and the HSRC with which MUT has a rising collaboration. As a Visiting International Research Fellow at the Africa Institute of South Africa of the HSRC, MUT grabbed the opportunity to have Dr Inyang who brings selected work experience credentials from the United Nations Commission on Sustainable Development, former Chairperson of the Science Advisory Board (Environmental Engineering Committee) of USEPA in Washington DC, USA. He is a former President of the African University of Science and Technology, Abuja and former Vice Chancellor of the Botswana International University of Science and Technology. The hybrid webinar was attended by academics.



Dr Hilary Inyang

In his presentation, Dr Inyang notes that "mining of materials has generated both gems and wastes that have added to the huge inventory of waste materials produced through civil and industrial infrastructure development in basically every sector of the economy: energy production, manufacturing, transportation, civil construction, domestic activities and agriculture. The wealth, stealth and health of countries have depended on the extent to which they can use knowledge systems to exploit and/or convert materials in processes that have generated wastes as by-products. With increase in population and waste production per capita, the tolerance of the environment and waste management capacity have both been seriously challenged with negative impacts on both the environment and human health. Increasing pollution of air, land and water resources threatens the sustainability of many communities, especially in Africa where systems are inadequate for cost-effective management of such negative impacts."

Dr Inyang also put emphasis that "among the environmental stressors in Africa are generated industrial and domestic wastes, emissions from hydrocarbon-burning systems, dust generation and transport, invasive species and transient hydrological events. The negative impacts of these environmental stressors are exacerbated by global climate change which as estimated by the Intergovernmental Panel on Climate Change (IPCC), will increase average annual global temperature by 1.8 to 4.0 degrees beyond pre-industrial levels before the end of this Century. The increasing impacts of global climate change in Africa are increased seasonality of rainfall leading to droughts in some regions and flooding in others; erosion of land (including loss of soil nutrients and damage to physical infrastructure) and increase in pollution of surface water bodies; desertification; inundation of coastal facilities due to sea level rise; and relocation of pests. These environmental stressors coupled with the traditional types, have significant implications on the human health, environment, and livelihood of people in various regions of Africa."

Therefore, for "greening of the economy through both mitigation and adaptation to environmental stresses requires the use of many instruments that fall within the categories of regulations, policies, monitoring systems, research, technological systems, market incentives, enforcement and international treaties."

In response to Dr Inyang's presentation, Dr Manyane Makua, Acting DVC Teaching and Learning, expressed gratitude to Dr Inyang and the HSRC for having provided MUT with such an auspicious speaker. "The webinar could not have come at a better time for MUT and the people of Umlazi Township, who have witnessed first-hand experience of global warming during the KZN floods in April this year. For this reason, I challenge the organisers of this magnificent event to convene a major conference where experts in the league of Dr Inyang could present and provide solutions to climate change issues faced by the African diaspora and the world," said Dr Makua

These environmental stressors coupled with the traditional types, have significant implications on the **human health, environment, and livelihood of people in various regions of Africa.**



NEW COLLABORATIVE AGREEMENTS



MUT SIGNS AN MOU WITH A UNIVERSITY IN CAMEROON



Dr JeanGad Mukuna, with Dr Anette Mienie

In yet another ground up approach to show how the University values its internationalisation strategy, the Faculty of Engineering has seized an opportunity to partner with the University Institute of the Tropics (IUGET) - South Polytech School, in Cameroon. The departments of Electrical Engineering led by Dr Richard Chidzonga, Head of Electrical Engineering Department and Dr JeanGad Mukuna, Acting Head of Mechanical Engineering Department have already commenced discussions that will ensure that MUT students are a part of this great ambition of internationalisation which is spearheaded by the MUT Vice-Chancellor.

These two departments are currently working on an implementation plan that will see MUT receive the first cohort of students from Cameroon and ensure that

both universities become the centres of knowledge production. The team from Cameroon have placed great faith on MUT having looked at the curriculum that MUT offers as well as taking note that all MUT Engineering qualifications are accredited by the Engineering Council of South Africa (ECSA).

Academics from both MUT and Cameroon will provide either physical learning or Virtual Learning between MUT and Cameroon. "We are getting deeper into the African continent, and this will be a wake-up call to our students as they will get exposed to how their counterparts from the other side of Africa are optimising knowledge grasped as an investment into the future", said Mbali Mkhize, senior director, Marketing and Communications.

MUT SIGNS AN MOU WITH ESAIP

On 15 July 2022, Professor Christophe Rouvrais, Dean and CEO of ESAIP in Angers, France, penned a letter to Professor Marcus Ramogale, MUT Vice-Chancellor welcoming and expressing an interest in working with "your pedagogical team within the world of teaching and research. As Dean and CEO of ESAIP Graduate School of Engineering, I have the pleasure to contact you after the meeting between your team and our international department. The delegation representatives have confirmed your interest in developing agreements with higher education institutions in France to further enhance your international relations. We host on campus more than 25 different nationalities and the African students represent more than half of the international students."

This is what Professor Ramogale has been advocating, an internationalised student body and workforce. "Once you internationalise your university, everything falls into



place", a statement which Professor Ramogale can really relate to. As a young PhD graduate from Nottingham University a few moons ago, his internationalisation exposure accelerated his career at a young age.

MUT's partnership with ESAIP will take MUT to another level. ESAIP is a 30-year bilingual Graduate School of Engineering in Angers and Aix-en-Provence, accredited by CTI (French commission of engineering titles) and member of the prestigious "Conférence des Grandes Ecoles". ESAIP has trained more than 4 500 alumni in the fields of computer science and networks, risk, health,

safety, and environmental management. ESAIP is a national group with nine campuses in France. It is also part of a network of more than 91 partner universities all around the world.

Welcome to the international world MUT. Praise goes to Dr Makua who was tasked with leading the delegation and encouraged Mbali Mkhize to take a 300 km journey from Paris to represent MUT.

"MUT's signing of the MoU with ESAIP is the commencement of a new chapter for MUT. We believe that this exposure will help transfer knowledge to MUT and the vision that Professor Marcus Ramogale has about entrepreneurship to be enhanced. It is no small matter that our students would have exposure to international students from more than 25 countries. From a teaching and learning perspective which is my forte, I am exhilarated, and I know that our university will take the stature that it deserves", said Dr Makua.

SASUF VISITS MUT



Helin Backman Kartal, seated, left, with MUT and University of Zululand staff

introduce SASUF to MUT and to show the benefits that MUT would have as a member of SASUF. Helin Backman Kartal, the SASUF Coordinator based at Uppsala University in Sweden was accompanied by the SASUF student leader and academics from the University of Zululand (Unizulu) who have played a pivotal role in strengthening Unizulu's collaborations with universities in Sweden.

Soon after the meeting with the Swedish representative, Professor Ramogale received an invitation, together with other South African Vice-Chancellors, to participate in a joint programme to deepen the collaboration between Swedish and South African universities.

The SASUF fever has hit MUT and the positive welcome of SASUF by MUT shows that "the internationalisation trajectory is moving very fast, and without leadership from the VC, we would not be accomplishing all the milestones outlined in the MUT Strategy 2025 so fast. Being part of SASUF has now also made us look around our region and be able to select partners who could collaborate with us in achieving the UN's 17 Sustainable Development Goals", said Mbali Mkhize, senior director, marketing, and communications.



On 29 July 2022, a delegation from Sweden landed at MUT to court the University to become a member of the South Africa-Sweden Forum (SASUF). The purpose of the meeting was to

MUT JOINS SOUTH AFRICA HIGHER EDUCATION WEEK

MUT staff in France

MUT seized an opportunity of a lifetime by joining the South Africa Higher Education (SA-HE) Week which was held in Paris from 27 June to 02 July. The week was filled with seminars, visits and engagements which were attended by high-powered decision-makers from SA and France's higher education sectors, ambassadors from both countries and major role players in the university and research sectors of both countries. The MUT delegation that was led by the DVC T&L (Acting) included Professor Babatunde Bakare, Dr Njabulo Gumede, Mbali Mkhize and Xolile Ngubane.

For the first time, MUT had an opportunity to become a part of the 23 universities from SA represented by 130 delegates and France was represented by 40 universities and various research organisations.

There is a strong bilateral cooperation between SA and France in Higher Education, Science, and Innovation. Currently, there are 1 500 bilateral cooperations under the Protea project. These achievements have been made possible by France institutions collaborating with universities in SA such as University of Cape Town, University of Pretoria, University of Johannesburg, University of the Witwatersrand, Nelson Mandela University, University of the Western Cape, Tshwane University Technology, and Stellenbosch University. Unfortunately, MUT had previously been excluded. The transformation taking place in France seeks to ensure more cooperation in Science, Technology, and Innovation.

Attending the SA Research Week enabled MUT to gain recognition and to have an opportunity for one-on-one engagements with prospective partners so that MUT could be well positioned to enter French universities and leverage multilateral cooperation and networks, including research infrastructure partnerships.

Lessons learnt during these visits are that French universities uplift one another as they rise, they empower one another and collaborate as a consortium, and they do not compete against one another which is what is common in many countries. In addition, South African universities began to acknowledge that a more inclusive approach is required internally to drive the research output of the country.

To optimise on being in France for a week, Dr Makua, leader of the delegation ensured that all team members got deployed to universities that had expressed interest in partnering with MUT. This saw the team crisscrossing France from Paris, Angers, Evry Courcouronnes, Montpellier and Marsellies all in one week.

DESTINATION ISIPCA SCHOOL (VERSAILLES FRANCE) MUT COLLABORATION

Professor Babatunde Bakare, Dr Njabulo Gumede and Xolile Ngubane were invited to view the cosmetics curriculum at the ISIPCA School of Cosmetics. The team, working with Dr Makua, Acting DVC Teaching and Learning is currently assembling the strengths of various departments such as chemistry, chemical engineering and the TCS as it prepares to draft a MoU between MUT and French cosmetic institutions, the first of these being ISIPCA School (Versailles, France).

The ISIPCA is a prestigious institution offering vocational education located in Versailles in France. This institution has two campuses, located in Versailles and Paris. This institution boasts world-class facilities to train students and industry professionals in fragrance, cosmetics, and food flavours.

Through the staff and student exchange, the partnership envisages to develop the competencies of staff in the Departments of Chemistry and Chemical Engineering and the TSC. Students from chemical and biological Sciences, botany (*Nature Conservation for MUT*), and chemical engineering will benefit through the collaboration by enrolling in a Master's programme in a variety of fields in cosmetics, fragrances and food flavours. This will improve the absorption of these students by the local cosmetic and food industries.

Curriculum Development

The ISIPCA collaboration presents unlimited opportunities for the development of a Postgraduate Diploma programme in chemistry and chemical engineering, which will consist of modules that will induct students in cosmetics, fragrance and food flavours. The introduction of the modules should then lead to a fully-fledged Cosmetic Science/Cosmetics Formulation diploma in future at MUT.

The areas for course development include but are not limited to:

- Scientific Principles of Cosmetic Formulations,
- Cosmetic Formulation Technology,
- Hair and Skin Biology for the Cosmetic Formulator,
- Cosmetic: Claims, Regulations and Ethics,



Professor Babatunde Barake, left, with Xoli Ngubane of the Technology Station in Chemicals

ISIPCA

- P A R I S -

- Professional Communication and Project Management for Cosmetic Scientists.

The ISIPCA collaboration will also benefit SMMEs through:

- Repackaging the existing TSC Cosmetic manufacturing manual so that it can be accredited as an NQF level 1 training course,
- The development of a training programme that is at NQF level 2, 3, 4 or 5, targeting entrepreneurs in the cosmetic space as well as scientists and engineers in the industry,
- Developing TSC staff expertise in Cosmetic formulations and fragrances, and
- Industry webinars.

The SMMEs will sign a Master Service Agreement (MSA) with the university to offer the above services. The MSA will also detail how the IP derived from the services rendered by MUT will be shared between TSC and the SMMEs. The university will determine the fee structure. Furthermore, the TSC will derive revenue from this incubation space that can be rented to SMMEs and from the services explained above.

SASUF GOES DIGITAL



SASUF GOES DIGITAL CONFERENCE A SPRINGBOARD FOR MUT'S INTERNATIONALISATION AGENDA

This is a link to MUT presenters: <https://sasufgoesdigital2022.sched.com/> and below are summaries of projects presented by MUT staff during the SASUF Goes Digital conference held from 19 to 23 September 2022.

STUDENT PERSPECTIVES ON GENDER-BASED VIOLENCE: GBV POSTER DESIGN AS A TOOL FOR PROMOTING STUDENT AGENCY AND SOCIAL JUSTICE

Dr Paulette Naidoo, Director: Student Counselling



Dr Paulette Naidoo

Gender-based Violence (GBV) compromises student mental health, holistic wellbeing, and academic success, ultimately affecting students' ability to graduate and contribute meaningfully to the economy, communities and societies at large. GBV intervention programmes that are relevant and responsive to students' realities are a necessary imperative in higher education. GBV posters play a pivotal role in GBV awareness campaigns as they succinctly integrate verbal and visual material to maximize the effect. The impact of GBV posters may be enhanced by affording students the autonomy and creative platform to design GBV posters which are reflective of their understanding and perceptions of GBV, as well as how they envisage effective justice and change.

SASUF GOES DIGITAL

19 - 23 SEP
#SASUF2030



THE MENTAL WELLBEING OF RURAL PEOPLE IN RELATION TO WATER AND FOOD SECURITY



Dr Thobeka Makhathini, left, with Dr Xolile Mkhize

Dr Xolile Mkhize, a Senior Lecturer in the Department of Community Extension; Dr Yvonne Mvuyane, Head of Department for Public Administration and Economics; and Dr Thobeka Makhathini, a Lecturer in the Department of Chemical Engineering were also joined by Fatima Peers and her team from the University of Johannesburg and the University of Venda.

According to these researchers, "climate change is set to affect marginalised communities in the global south and north the greatest. The loss of life, arable land, infrastructure, property, livelihoods, livestock and crops exacerbate rural people's vulnerability to poverty, unemployment, displacement and strains their precarious mental health."

FRACTIONATION AND EVALUATION OF CONTAMINATED SOILS FROM WASTEWATER DISPOSAL AND BURNING OF A CHEMICAL STORAGE FACILITY DURING THE JULY 2021 UNREST IN SOUTH AFRICA



Awonke Mbangi

Awonke Mbangi, a Lecturer in the Department of Agriculture.

Communities surrounding the UPL Limited (formerly United Phosphorus Limited) warehouse that was set

alight on 12 July 2021 during the violent looting that occurred in South Africa, have been contaminated with a myriad of pollutants that were kept at the warehouse. These chemicals are believed to have spread in two ways. The first was the toxic smoke caused by the fire, and the chemical concoction caused by the water used to fight the fire. This water flowed down the hill as it found the path of least resistance into a wetland and the Ohlanga River of the eThekweni Municipality.

A second site Darvill Waste Water Works (DWWW),

situated east of Pietermaritzburg, receives both domestic sludge from surrounding residences and industrial waste. Over 50 years of sludge application on about 57 hectares of land by the Darvill Waste Water Treatment Works has resulted in soil pollution by a myriad of heavy metals. It is the objective of the researchers to determine the extent of this pollution through fractionation and mapping as well as to raise awareness among the surrounding communities and screen phytoremediation as a method to clean up the contamination.

TRANSLANGUAGING AS A RESOURCE TO PROMOTE SOCIAL JUSTICE IN HIGHER EDUCATION



Professor Sandiso Ngcobo

Professor Sandiso Ngcobo, Head of the Department of Communication.

The presentation aimed to share that South Africa is not only a multilingual country because of the different languages spoken by its citizens but that the country has throughout the years attracted many nationalities from mainly the African continent and across other parts of the world. This brings attention to the need for the development of multilingualism in education at different levels.

TRANSLATING LESSONS LEARNT FROM COVID-19



Professor Sandiso Ngcobo

Xolile Ngubane Acting Director of the Technology Station in Chemicals (Gender mainstreaming, youth, Covid-19 and rural entrepreneurship).

This presentation which was anchored by professor Thuli Madonsela and featuring Ngubane showed that to reintegrate the family, poor people particularly in the rural areas, need to have strong family households. Drawing from lessons learnt during Covid-19, the presentation revealed that before and during Covid 19, diversity-blind policies that amplify inequality, especially for the poor and women were developed and enforced. The workshop was a preparation for an upcoming five-week PhD course syllabus on Rural Development and Social Justice. The proposed course is a collaboration between Stellenbosch University, MUT, Swedish University of Agricultural Sciences, Stockholm University, and Södertorn University.

BUILDING SUSTAINABLE CITIES



Dr Charlotte Bongekile Yvonne Mvuyane

Dr Yvonne Mvuyane, Head of Department for Public Administration and EconomicsDr Bheka Ntshangase, Acting Dean, Management Sciences, and Andile Mafa, MUT Alumnus

According to the researchers, “cities in the world are regarded as a gateway for commerce, social, human and economic development. To ensure that developments

take place, cities should prioritise amongst others spatial planning, integrated transport system, water management systems, proper sanitation, and disaster risk reduction to achieve sustainable human development”.

The project focuses on the challenges faced by local governments/sub-national in building resilient and sustainable cities to achieve sustainable human development. This is in line with SDG 11.

A NEED FOR MENTAL HEALTH SUPPORT SERVICES IN UNIVERSITIES



Arthi Ramrung

Arthi Ramrung, Lecturer in the Teaching and Learning Development Centre partnered with the University of the Witwatersrand and the University of the Western Cape.

The experience of South African students in higher education is uneven, due to historical imbalances and the lack of support needed to be successful at

university. Studies conducted across HE in SA show that students transitioning from under-resourced secondary schools are generally underprepared for tertiary studies; most are first-generation students from previously disadvantaged backgrounds. Massification of HE, along with these factors contributes to concerning throughput rates, high attrition rates, and a reduced sense of belonging. Similarly, the rapid increase of postgraduate students in the sector poses challenges in terms of postgraduate pedagogies, supervision capacity, support for postgraduate students, and throughput rates.

BUILDING RESILIENCE AMONG DISADVANTAGED COMMUNITIES



Mbali Mkhize

Mbali Mkhize, Senior Director: Marketing and Communication and Zama Sishi, Director: Stakeholder Relations Management; partnering with the University of Venda, Gift of the Givers, Environmental and Language Trust, Mina Foundation and the voice of students and young people represented by Lisa Mbongwa, Marketing and Communication Administrator.

The project aimed to show that during disasters, communities wait for help not realising that help cannot even reach them in time because their areas might be inaccessible. The workshop consisted of teams from multi-stakeholder groupings with the purpose of building and empowering communities in the event of future disasters.

The project is now a part of a multi-university consortium with five universities across SA and Sweden jointly developing a project around this concept.

SKILLS DEVELOPMENT AND EQUITABLE EDUCATION



Professor Busisiwe Nkonki-Mandleni

Professor Busisiwe Nkonki-Mandleni, Director: Community Engagement and Development Directorate.

The South African education system has been greatly challenged by the lack of equality that has compromised the quality of education. This has resulted in adverse consequences which include:

- Inadequately equipped and sometimes no equipment in early childhood centres to stimulate children at their initial stages of growth;
- Uninformed high school learners about the correct career choices;
- Lack of entrepreneurial mindset;
- Illiterate parents; and
- Low pass rates, increasing rate of matric drop-outs, and a shortage of qualified teachers.

THE ROLE OF CONVOCATION AND ALUMNI IN THE GROWTH TRAJECTORY OF THE UNIVERSITY, WITH SPECIFIC REFERENCE TO UNIVERSITIES OF TECHNOLOGY

Dr Connie Israel, Executive Director: Institutional Advancement.

The presentation aimed to share and discuss the traditional role of convocation and alumni associations to their alma mater, how the universities could build a formal relationship with alumni, and how they may shape these through online communicative platforms.

OPPORTUNITY OF A LIFETIME FOR ZININGI JAYA



Nobuhle Ziningi Jaya

science to address health issues in our communities”, said Ziningi.

Attending this conference helped Ziningi to participate in science communication training. Having this skill will help her organise workshops and training sessions for colleagues to learn different ways in which research findings can be communicated to the general population using methods other than journal publications and conference presentations.

Ms Ziningi said “I will play an active role in collaborating with various colleagues and international partners to bring awareness and education about climate change and the small lifestyle changes we can make to reduce or slow down climate change. The most common trajectory of careers in research is that of being in academia or of working as a researcher. However, after attending the conference it became clear that researchers are no longer limited to just these options, a science career in politics has also become an option. Although I am not sure whether such is recognised in South Africa, I would like to present it as an option for many of my students and colleagues who are open to working abroad. I will do more research on this to present complete information to those who are interested”.

Nobuhle Ziningi Jaya an nGap lecturer in the faculty of Natural Sciences' Biomedical Health Sciences received the opportunity of a lifetime to attend the EuroScience Open Forum (ESOF) 2022 which took place on 13 to 16 July 2022 in Leiden (Netherlands), the European City of Science 2022. As an nGap lecturer, she is part of the New Generation of Academics Programme (nGAP), a prestigious programme under the DHET which involves the recruitment of highly capable scholars as new academics.

Ziningi was afforded a travel grant by the Department of Science and Innovation in response to a call for an expression of interest. The main theme of the 10th edition of ESOF2022 was “Crossing Borders, Engaged Science, Resilient Societies”. “The conference comprised of seven different themes and I submitted an application for one of the themes which was “freedom and responsibility of science”. I was interested in this theme because of my research interest which involves interventions that marry public health and biomedical

The conference provided her with networking and relationship building opportunities with scientists in the STEM field. “I had the opportunity to meet with scientists who specialise in technological advancements in the laboratory, and who facilitate learning including e-learning pedagogy. This exposure will help strengthen research activities in the biomedical sciences as well as teaching and learning practices particularly e-learning pedagogical practices.”